

Design Note: Architecture & Low-Level Design of the LLD Practice Platform

Author: LLD Practice Platform Team **Date:** September 2026 **Assignment:** CipherSchools Engineering Hiring Assignment **Focus:** MVP Definition, Domain Architecture, Design Patterns, Trade-offs & Extensibility

1. Executive Summary & MVP Scope

The **LLD Practice Platform** is engineered as a focused, modular monolith designed around a single core learner problem: providing objective, rubric-grounded, explainable design feedback that powers iterative improvement.

In accordance with good engineering judgment, we deliberately reject bloated peripheral systems (such as multi-tenant authentication, collaborative diagram whiteboards, or distributed microservice orchestration) in favor of deep, cohesive domain modeling.

1.1 Scope Boundaries

CapabilityIn Scope (MVP)Out of Scope (Deferred with Rationale) :

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Problem Catalog5 curated LLD problems (Parking Lot, Elevator Dispatcher, Splitwise, Cache System with LRU/LFU/FIFO, API Rate Limiter) with business invariants and rubrics.Crowdsourced problem creation and uncuration community voting. **Submission Format**Structured design specification (Core Entities, Responsibilities, Interfaces, Patterns, Trade-offs).Heavy free-form graphical drag-and-drop vector drawing tools. **Visual & Code Bridge**Dynamic Live UML class diagram synthesis and multi-language boilerplate code generator (Java, TypeScript, C++).Full in-browser compiler sandbox or multi-language execution runtime. **Evaluation Engine**Hybrid two-stage evaluation: Deterministic Rule Evaluator + Rubric Evaluator with concrete evidence citations.Heavy distributed message broker (Kafka/RabbitMQ); modular in-process state transitions are sufficient for MVP. **Feedback Model**Multi-dimensional rubric feedback citing concrete evidence, design concerns, and remedies.Vague, single-prompt subjective AI grading ("Looks good! 85/100"). **Attempt Tracking**Versioned attempt history (\$A_1, A_2, \dots\$) with rubric delta comparison and score progression.Social leaderboards, public user profiles, or gamification badges.

2. Learner User Flow

The platform guides the learner through an active feedback-driven state cycle:

```

mermaid stateDiagram-v2 [*] --> BrowseCatalog: Learner arrives BrowseCatalog -->
SelectProblem: Selects curated LLD problem (e.g. Parking Lot, Cache) SelectProblem -->
CreateAttempt: Initiates Attempt #1 (DRAFT) CreateAttempt --> DraftDesign: Fills structured
specification DraftDesign --> InspectUML: Real-time Live UML diagram preview DraftDesign -->
ExportCode: Generate Java / TypeScript / C++ skeleton DraftDesign --> SubmitSolution: Submits
design state "Attempt Lifecycle" as AttemptLifecycle { SubmitSolution --> SUBMITTED: Persist
payload safely SUBMITTED --> EVALUATING: Trigger evaluation pipeline EVALUATING --> COMPLETED:
Deterministic + Rubric Success EVALUATING --> FAILED: Evaluator timeout / error FAILED -->
EVALUATING: Retry evaluation } COMPLETED --> ReviewFeedback: Inspect Evidence, Concerns,
Suggestions ReviewFeedback --> ViewHistory: Compare with prior attempts & rubric delta
ReviewFeedback --> CreateAttempt: Try Again (Initiates Attempt #2)

```

3. Domain Model & Object-Oriented Design

The core domain adheres strictly to SOLID principles, DDD (Domain-Driven Design) layering, and clean separation between domain entities, value objects, and evaluation strategies.

3.1 Domain Class Diagram

```

mermaid classDiagram
class Problem { +ProblemId id +String title +String description
+List~String~ functionalRequirements +List~String~ nonFunctionalRequirements +Rubric rubric
+Template starterTemplate +getRubric() Rubric }

class Attempt { +AttemptId id +ProblemId problemId +String learnerId +int attemptNumber
+AttemptStatus status +DateTime createdAt +DateTime updatedAt +Submission submission +Feedback
feedback +submit(Submission submission) void +startEvaluation() void
+completeEvaluation(Feedback feedback) void +failEvaluation(String reason) void }

class AttemptStatus { <> DRAFT SUBMITTED EVALUATING COMPLETED FAILED }

class Submission { +SubmissionId id +SubmissionFormat format +ISubmissionPayload payload
+DateTime submittedAt +getPayload() ISubmissionPayload }

class ISubmissionPayload { <> +validate() ValidationResult +getSummary() String }

class StructuredDesignPayload { +List~EntitySpec~ entities +List~InterfaceSpec~ interfaces
+List~PatternSpec~ designPatterns +String tradeOffsAndAssumptions +validate() ValidationResult
+getSummary() String }

class Rubric { +RubricId id +List~RubricCriterion~ criteria +getMaxScore() int }

class RubricCriterion { +String id +String name +String description +int weight +int maxScore }

class Feedback { +FeedbackId id +int overallScore +List~CriterionEvaluation~ evaluations +String
summary +DateTime evaluatedAt }

class CriterionEvaluation { +String criterionId +int score +int maxScore +String evidence
+String concern +String suggestion +float confidence }

class IEvaluator { <> +evaluate(Submission submission, Rubric rubric) Promise~EvaluationResult~

```

```

}

class DeterministicRuleEvaluator { +evaluate(Submission submission, Rubric rubric)
Promise~EvaluationResult~ }

class SemanticRubricEvaluator { +evaluate(Submission submission, Rubric rubric)
Promise~EvaluationResult~ }

class CompositeEvaluator { -List~IEvaluator~ evaluators +evaluate(Submission submission, Rubric
rubric) Promise~EvaluationResult~ }

Problem "1" *-- "1" Rubric : specifies Attempt "1" *-- "0..1" Submission : contains Attempt "1"
*-- "0..1" Feedback : produces Attempt --> AttemptStatus : tracks Submission "1" *-- "1"
ISubmissionPayload : encapsulates StructuredDesignPayload ..|> ISubmissionPayload : implements
Feedback "1" *-- "many" CriterionEvaluation : details CompositeEvaluator ..|> IEvaluator :
implements DeterministicRuleEvaluator ..|> IEvaluator : implements SemanticRubricEvaluator ..|>
IEvaluator : implements CompositeEvaluator "1" o-- "many" IEvaluator : coordinates `

```

4. Key Responsibilities of Domain Classes

Class / Interface Primary Responsibility Justification & SOLID Adherence :

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Problem Encapsulates problem statements, constraints, and rubric definitions. **Single Responsibility Principle (SRP)**: Holds domain problem invariants; does not know about user attempts or evaluation mechanics. **Attempt** Manages the lifecycle state machine of a learner's solution effort. **Encapsulation & State Invariants**: Enforces legal state transitions (DRAFT $\rightarrow$ SUBMITTED $\rightarrow$ EVALUATING $\rightarrow$ COMPLETED). Guarantees an attempt cannot be marked completed without valid feedback. **Submission** Immutably captures what the learner submitted at a point in time. **Open-Closed Principle (OCP)**: Uses generic ISubmissionPayload to decouple the attempt from the format of the solution. **Rubric & RubricCriterion** Models the multi-dimensional criteria against which designs are judged. Decouples grading rules from both problems and evaluators. Rubrics can be versioned or customized per problem type. **IEvaluator** Strategy contract for evaluating a submission against a rubric. **Dependency Inversion Principle (DIP)**: Application services depend on the IEvaluator abstraction, not concrete AI or rule engines. **CompositeEvaluator** Coordinates deterministic checks with semantic AI evaluations. **Composite & Open-Closed Patterns**: Allows chaining any number of evaluators without altering the calling service.

4.1 Visual UML Synthesis & Polyglot Code Generation

To prevent the "passive reading trap" and reinforce the bridge between design thinking and real-

world implementation, the platform incorporates dual synthesis engines:

- * **Dynamic Live UML Synthesis:** Translates the learner's declared entities, responsibilities, and interfaces into interactive visual class nodes and standard Mermaid.js class diagrams in real time.
- * **Polyglot Code Exporter:** Automatically translates the structured design specification into strongly typed, compilable boilerplate across **Java** (interfaces, immutability, thread-safe patterns), **TypeScript** (typed interfaces, ES6 classes), and **C++** (abstract base classes, virtual destructors, modern C++20 pointers).

5. Evaluation & Feedback Approach

Evaluation is structured into a two-phase pipeline to maximize reliability and speed while minimizing costs:

```
graph LR
    LR[LR Sub[Learner Submission]] --> Comp[Composite Evaluator]
    subgraph Phase_1 [Phase 1: Deterministic Engine <50ms]
        Comp --> DET[DET[Deterministic Rule Evaluator]]
        DET --> D1{D1{Schema & Completeness Valid?}}
        D1 -- No --> FailFast[Fail Fast: Highlight Missing Sections]
        D1 -- Yes --> RuleScore[RuleScore[Calculate Structural Score]]
    end
    subgraph Phase_2 [Phase 2: Semantic Rubric Engine]
        RuleScore --> LLM[LLM[Semantic Rubric Evaluator]]
        LLM --> Prompt[Prompt[Inject Structured Rubric & Evidence Anchor]]
        Prompt --> Parse[Parse[Validate Structured JSON Response]]
    end
    Parse --> Merge[Merge[Synthesize Overall Feedback]]
    FailFast --> Merge
    Merge --> Res[Res[Return Unified Feedback]]
```

5.1 Deterministic Checks

- * **Section Coverage:** Ensures all mandatory components (Entities, Responsibilities, Interfaces, Pattern justifications, Assumptions) are present.
- * **Relationship Integrity:** Verifies that referenced interfaces and classes are actually declared.
- * **Minimum Elaboration Depth:** Flags placeholder or one-word descriptions before wasting LLM tokens.

5.2 Semantic Checks (Rubric-Grounded)

The semantic engine does not ask the LLM *"Is this design good?"* (which results in inconsistent grades). Instead, it evaluates 4 strict orthogonal dimensions:

1. **Domain Modeling & Completeness:** Did the design capture all core entities required by the problem?
2. **Single Responsibility & Cohesion (SRP):** Are class responsibilities focused, or do "God objects" exist?
3. **Coupling & Extensibility (OCP / DIP):** Are algorithms and behaviors abstracted behind interfaces (e.g. Strategy/Factory patterns)?
4. **Trade-offs & Constraints:** Did the candidate explain why they chose their design and what trade-offs were accepted?

5.3 Feedback Shape

Every criterion evaluation adheres to a strict contract:

- * **criterionId** : Identifier matching the problem rubric.
- * **score** : 1 to 5.
- * **evidence** : Explicit citation of class/method names from the candidate's submission.
- * **concern** : Concrete failure scenario or design smell.
- * **suggestion** : Specific pattern or refactoring to address in the next attempt.
- * **confidence** : Evaluator certainty score (0.0 - 1.0).

6. The Two Change Tests

A critical test of domain design is how gracefully it absorbs change. Our architecture was specifically tested against the two scenarios highlighted in the hiring brief:

Change Test A: Supporting a New Submission Format (e.g., UML Class Diagram or Code)

*** The Challenge:** Today the platform accepts structured markdown/text. Tomorrow it must support a UML Class Diagram or raw Java/TypeScript code. *** How Our Architecture Absorbs It:** 1. We defined `ISubmissionPayload` : ``typescript export interface ISubmissionPayload { format: SubmissionFormat; // 'STRUCTURED_TEXT' 'UML_DIAGRAM' 'CODE' validate(): ValidationResult; toEvaluationContext(): string; }`` 2. To support UML diagrams, we implement `UmlDiagramPayload` implements `ISubmissionPayload` containing diagram nodes/edges or Mermaid/PlantUML syntax. 3. Attempt , `PracticeService` , and the database schema remain **completely unchanged**. The `Submission` entity wraps any payload adhering to `ISubmissionPayload` .

Change Test B: Adding a New Evaluator (e.g., Static AST Linter or Human Review)

*** The Challenge:** Today feedback is generated by automated rules and an LLM. Tomorrow we add a rule-based AST linter or asynchronous human peer review. *** How Our Architecture Absorbs It:** 1. Evaluators implement `IEvaluator` : ``typescript export interface IEvaluator { readonly name: string; evaluate(submission: Submission, rubric: Rubric): Promise; }`` 2. To add human review, we simply create `HumanReviewEvaluator` implements `IEvaluator` . 3. We register new `HumanReviewEvaluator()` into `CompositeEvaluator` : ``typescript const evaluator = new CompositeEvaluator([ new DeterministicRuleEvaluator(), new SemanticRubricEvaluator(), new HumanReviewEvaluator() // <-- Plugs in seamlessly! ]);`` 4. The practice flow (`PracticeService.submitSolution()`) does not change by a single line. It simply invokes `this.evaluator.evaluate()` .

7. Practical Scale & Architectural Trade-offs

DecisionChosen ApproachAlternative ConsideredRationale & Trade-off :

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System ArchitectureModular MonolithMicroservices (Service per feature)For a practice platform prototype, a clean monolith eliminates network serialization, distributed transactions, and

deployment complexity while maintaining strict domain boundaries. **Evaluation Timing** Asynchronous **Lifecycle with Immediate Save** Synchronous blocking HTTP request LLM evaluation takes 2-5 seconds. Blocking the HTTP connection risks browser timeouts and loses submissions if the network drops. We save the attempt as **SUBMITTED** first, then transition to **EVALUATING** asynchronously. **State Storage** **Domain Repository Interface (In-Memory + File Persistence)** Distributed Database (PostgreSQL / DynamoDB) Adheres to Dependency Inversion. The repository interface (**IAttemptRepository**) allows switching to a cloud database without touching domain logic, while keeping the prototype zero-setup for reviewers. **Idempotency & Retries** **Attempt-Level State Machine** Uncontrolled client re-requests If a learner clicks "Submit" twice, the state machine recognizes **SUBMITTED** or **EVALUATING** and rejects duplicate evaluation jobs. If an evaluation fails, the attempt transitions to **FAILED** and can be retried without re-typing.

Component Separation Roadmap

If the platform scales to 50,000 daily active learners, the **first and only component to extract** is the **Evaluation Worker Pool**: * The web monolith continues handling problems, attempt management, and submission ingestion. * When a submission is saved, a lightweight event (**SubmissionCreatedEvent**) is published to a job queue (e.g., AWS SQS or Redis Streams). * Independent evaluation worker nodes consume from the queue, execute **CompositeEvaluator** , and write the resulting **Feedback** back to the database. * The domain entities (**Attempt** , **Submission** , **Rubric** , **Feedback**) remain identical.